

Environmental Fingerprint × Machine Learning Breeding Model for Winter Wheat – Summer Maize Rotation System in Pinggu District, Beijing

Preliminary Outline — For Advisor & Coworker Review

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1 Project Overview

This project develops a machine learning framework that integrates environmental fingerprints—comprehensive gridded climate, soil, and topographic variables—with multi-variety crop traits and management scenarios to predict yield and optimize agronomic practices. The study area is Pinggu District, Beijing (234 grids at 1 km resolution), operating under a winter wheat – summer maize double-cropping rotation system.

The core innovation lies in moving beyond traditional site-specific or single-factor analysis to a spatially explicit, multi-dimensional approach: environmental fingerprints capture the full spectrum of biophysical conditions at each grid cell, while machine learning models (XGBoost, Random Forest, Quantile Random Forest) decode the complex genotype × environment × management (G×E×M) interactions. Model interpretation via SHAP (SHapley Additive exPlanations) values, combined with Bootstrap confidence intervals, provides both mechanistic insight and uncertainty quantification.

2 Research Background & Knowledge Gap

Crop yield prediction has traditionally relied on process-based models (e.g., APSIM, DSSAT) that require extensive parameterization, or statistical models that lack spatial transferability. Recent advances in

machine learning offer an alternative: data-driven models can capture non-linear $G \times E \times M$ interactions without explicit physiological equations. However, three critical gaps remain:

- Environmental representation: Most ML-based yield studies use a handful of climate variables (e.g., seasonal mean temperature and precipitation), losing information on extremes, seasonality, and soil constraints.
- Uncertainty quantification: Point predictions dominate the literature; prediction intervals and spatial uncertainty maps are rare.
- Management optimization under environmental constraints: Existing optimization studies (e.g., Xiao et al. 2024, Nature Food) use aggregated regional data; grid-level, multi-objective optimization coupling environmental fingerprints with ML is unexplored.

This project addresses all three gaps simultaneously.

3 Study Area & Cropping System

Parameter	Value
Location	Pinggu District, Beijing (117.1° E, 40.1° N)
Grid resolution	1 km × 1 km (234 grid cells)
Cropping system	Winter wheat (Oct – Jun) – Summer maize (Jun – Oct) rotation
Climate	Temperate monsoon; annual T 11 – 12° C; annual P 600 – 650 mm
Wheat varieties	8 varieties (Jingdong 8/12/17/22, Zhongmai 175/1062, Lunxuan 987/169)
Maize varieties	8 varieties (literature-extracted summer maize cultivars)
Management data	Self-designed 14 scenarios + Xiao et al. (2024) public dataset (CC BY 4.0)

4 Data Inventory

4.1 Environmental Fingerprint Matrix

The core dataset is a 234×73 matrix, where each row is a 1-km grid cell and each column is an environmental variable. All data have been extracted, quality-controlled, and compiled into a single CSV (Outputs/pinggu_environmental_data.csv). Missing data rate: 0.3% (3 terrain-derived variables at edge grids). Modelable variables: 71/73 (97%).

Category	Source	Variables	Resolution	Status
Bioclimatic	WorldClim 2.1	19 (biol - bio19)	2.5 arc-min	Ready
Monthly temperature	WorldClim 2.1	12 (tavg 01 - 12)	2.5 arc-min	Ready
Monthly precipitation	WorldClim 2.1	12 (prec 01 - 12)	2.5 arc-min	Ready
Soil properties	SoilGrids 2.0 (ISRIC)	8 (sand/silt/clay/SOC/N/pH/BD/CEC, 0 - 5 cm)	250 m $\rightarrow$ 5 km	Ready
Topography	SRTM 90m (CGIAR-CSI)	3 (elevation, slope, aspect)	90 m	Ready
Growing degree days	Derived	6 (GDD0/5/10/15/20/25)	1 km	Regenerated from tavg
Soil water constants	Derived	3 (SAT, DUL, LL15)	1 km	Regenerated from SoilGrids
Variety traits	Literature + approval docs	8 wheat $\times$ 8 traits; 8 maize $\times$ 9 traits	Variety-level	Ready
Management scenarios	Self-designed	14 scenarios (N $\times$ Irrigation $\times$ Density)	Grid-level	Ready
Management - external	Xiao et al. (2024) Figshare	4 files (N, irrigation, yield; baseline + SSPs)	1 km	Ready (CC BY 4.0)

4.2 Management Data: Two Complementary Sources

Source 1 (Self-designed): 14 management scenarios combining N fertilization (0/150/300 kg ha⁻¹), irrigation (0/200/400 mm), and sowing density adjustments, applied to all 234 grids via management_grid_default.csv. Designed following Chen et al. (2014) Nature and Bai et al. (2024) Nature Food 5:148 - 159.

Source 2 (Public): Xiao et al. (2024) "Spatiotemporal co-optimization of agricultural management practices" (Nature Food 5:59 - 71). Dataset downloaded from Figshare (DOI: 10.6084/m9.figshare.24471919.v5, CC BY 4.0)

license). Contains optimized N fertilizer, irrigation, and yield maps for wheat and maize across three time periods (baseline, 2030s, 2060s) under SSP2-4.5 and SSP5-8.5. Four key files downloaded (61 MB total); primary data file Allregions.rds (57.3 MB) requires R-to-CSV conversion.

5 Analytical Methods Pipeline

5.1 Feature Engineering (Stage R1 + Auditor S1)

The 73-variable environmental fingerprint matrix will undergo systematic feature engineering in five steps: (1) data cleaning (outlier handling, imputation), (2) feature construction (interaction terms, vegetation indices if available), (3) dimensionality assessment (PCA, variance inflation factor), (4) feature selection (Boruta, recursive feature elimination with SHAP), and (5) sensitivity analysis (one-at-a-time perturbation).

Feature auditor S1 independently validates R1 output using alternative methods (e.g., mutual information vs. Boruta) and flags discrepancies for discussion before the modeling stage.

5.2 Modeling (Stage R2 + Auditor S2)

The modeling pipeline follows a progressive complexity strategy:

Step	Method	Purpose	Expected Output
1. Baseline	Multiple Linear Regression	Establish lower performance bound	R ² , RMSE baseline
2. Ensemble trees	Random Forest (Breiman 2001)	Robust non-linear baseline	Feature importance ranking
3. Gradient boosting	XGBoost (Chen & Guestrin 2016)	State-of-the-art structured data model	Optimized via Optuna
4. Hyperparameter tuning	Optuna (Akiba et al. 2019)	Bayesian optimization of XGBoost params	Best trial + param landscape
5. Uncertainty quantification	Quantile Random Forest (Meinshausen 2006)	90% prediction intervals per grid	Spatial uncertainty maps
6. Multi-objective optimization	NSGA-II	Yield - N fertilizer Pareto frontier	Optimal management portfolios

Model validation uses spatial cross-validation (block k-fold, $k=5-10$) to prevent spatial autocorrelation from inflating performance estimates. Auditor S2 independently tests model robustness via permutation tests, residual spatial autocorrelation analysis (Moran's I), and sensitivity to hyperparameter perturbations.

5.3 Interpretation (Stage R3)

SHAP (Lundberg & Lee 2017) values decompose each prediction into additive feature contributions. Key analyses include: (1) global feature importance ranking with Bootstrap 95% confidence intervals, (2) SHAP dependence plots for the top 8–10 features, (3) SHAP interaction values to detect non-additive $G \times E$ effects, (4) spatial mapping of SHAP values to identify where each environmental factor dominates yield limitation, and (5) construction of an Environmental Effect Similarity Index (EESI) for recommending variety deployment zones.

5.4 Causal Inference (Optional Extension)

If time permits, the PCMRI algorithm (Runge et al. 2019) will be applied to the environmental fingerprint time series to distinguish direct causal drivers from indirect correlations, strengthening the mechanistic interpretation of SHAP results.

6 Innovation Points

Priority	Innovation	Novelty	Feasibility
P0	Dynamic environmental fingerprints: 73-variable gridded representation replacing traditional few-variable climate summaries	First application of full-spectrum environmental fingerprinting at 1-km resolution in North China Plain	Data already compiled
P0	Spatial cross-	Addresses spatial	Code available in

	validation with uncertainty: Block k-fold CV + QRF prediction intervals	autocorrelation inflation (common in agricultural ML) and provides per-grid confidence	scikit-learn/scikit-garden
P1	SHAP interaction-based G×E×M decomposition with Bootstrap confidence intervals	Moving SHAP from point estimates to statistical inference; directly interpretable by agronomists	Requires R/shapr or Python/shap
P1	Environmental Effect Similarity Index (EESI) for variety recommendation zoning	Translates black-box model output into actionable agronomic decisions	Concept proven; requires implementation
P2	Causal structure learning via PCMCi on environmental fingerprint series	Quantifies direct vs. indirect environmental effects; rare in agricultural ML	Algorithm available; depends on time availability
P2	Multi-objective NSGA-II optimization under environmental constraints	Grid-level management portfolios balancing yield, N use, and environmental cost	pymoo/deap libraries available

7 Team Structure & Workflow

The project adopts a dual-track validation architecture (3R + 5S, v4). Three primary roles (R1 - R3) execute the core pipeline; two audit roles (S1 - S2) independently verify each stage; one writing role (S4) operates throughout; and two visualization roles (S5 - S6) produce all figures. Every primary output is cross-validated by its auditor before advancing to the next stage.

Code	Role	Responsibility	Key Deliverable
R1	Feature Engineering Architect	Clean → Construct → Reduce → Select features	Final feature matrix (234 × k)
S1	Feature Auditor	Independently verify R1 with alternative methods	Audit report + approved feature set
R2	Modeling & Optimization Engineer	Baseline → Ensemble → XGBoost → QRF → NSGA-II	Trained models + predictions + Pareto fronts
S2	Model Validator	Spatial CV +	Validation report +

		permutation tests + Moran's I audit	approved models
R3	SHAP Interpretation Engineer	SHAP analysis + Bootstrap CI + EESI construction	SHAP package + interpretation report
S4	Manuscript Writer	Draft paper throughout pipeline (continuous)	Manuscript draft (target: JIA)
S5	Figure Designer (Main)	Design and produce Fig 1 - 7 (600 DPI TIFF)	7 main-text figures
S6	Figure Designer (Supp) + Style Manager	Design Fig S1 - S12 (300 DPI PNG); maintain matplotlib style	12 supplementary figures + .mplstyle

Estimated timeline: 17 working days (~3.5 calendar weeks), with a buffer for iteration and advisor review.

7.1 Data Flow

All source data reside in Data/ (WorldClim, SoilGrids, SRTM, Variety, Management). Feature engineering scripts in src/r1_feature_engineering/ read from Data/ and write intermediate outputs to Outputs/intermediate/. Modeling scripts in src/r2_modeling/ read the engineered feature matrix, train models, and save .pkl files to Outputs/models/. Interpretation scripts in src/r3_shap_interpretation/ load models and produce SHAP outputs. All figures are generated under Outputs/figures/ (main/ for S5, supp/ for S6) using the shared matplotlib style sheet at config/style_2026sp.mplstyle.

8 Expected Outcomes

8.1 Scientific Outputs

- One peer-reviewed manuscript targeting Journal of Integrative Agriculture (JIA, IF ~4.0), with Agronomy-Basel or PeerJ as fallback venues.
- A validated, interpretable ML framework for gridded yield prediction and management optimization applicable to other regions.

- An open-source codebase (GitHub: github.com/DearDragon233/2026-SP) containing all analysis scripts, organized by role module.

8.2 Figures (19 total)

Figure	Content	Type	Producer
Fig 1	Study area + environmental fingerprint design schematic	Overview	S5
Fig 2	Feature importance ranking with Bootstrap 95% CI	Methods/Results	S5
Fig 3	Model performance comparison (5 models, spatial CV)	Results	S5
Fig 4	Spatial yield prediction maps with QRF uncertainty	Results	S5
Fig 5	SHAP summary + dependence plots (top 8 features)	Results	S5
Fig 6	SHAP interaction heatmap + G×E effect decomposition	Results	S5
Fig 7	NSGA-II Pareto frontier + optimized management portfolios	Results	S5
Fig S1 - S12	Data diagnostics, sensitivity analyses, supplementary maps	Supplementary	S6

8.3 Data Products

- Cleaned environmental fingerprint matrix (234 grids $\times$ k features), publicly archived with DOI upon publication.
- Extracted Xiao et al. (2024) management data in CSV format for Python users (original .rds file retained).
- Trained model checkpoints (.pkl) for reproducibility.

9 Key References

The full reference guide (34 papers with detailed reading notes) is available in ManageFiles/26SP_参考文献精读与使用指南.pdf. Ten OA papers have been downloaded to Paper/. Core methodological references:

Chen T, Guestrin C. 2016. XGBoost: A scalable tree boosting system. Proceedings of KDD 2016. arXiv:1603.02754.

Lundberg SM, Lee SI. 2017. A unified approach to interpreting model predictions. NeurIPS 2017. arXiv:1705.07874.

Breiman L. 2001. Random forests. Machine Learning 45(1):5 – 32.

Meinshausen N. 2006. Quantile regression forests. Journal of Machine Learning Research 7:983 – 999.

Shahhosseini M et al. 2021. Coupling machine learning and crop modeling. Agricultural and Forest Meteorology 307:108486.

Xiao L et al. 2024. Spatiotemporal co-optimization of agricultural management practices. Nature Food 5:59 – 71. [Data: Figshare, CC BY 4.0]

Bai S et al. 2024. Environmental fingerprints for crop yield prediction. Nature Food 5:148 – 159.

Runge J et al. 2019. Detecting and quantifying causal associations. Science Advances 5:eaau4996.

Akiba T et al. 2019. Optuna: A next-generation hyperparameter optimization framework. KDD 2019. arXiv:1907.10902.

10 Current Status & Next Steps

Item	Status
Environmental fingerprint CSV (234 × 73)	Complete
Variety trait data (8 wheat + 8 maize)	Complete
Management scenarios (14 self-designed + Xiao 2024 public data)	Complete
Reference paper PDFs (10 OA papers, 17 MB)	Complete
Project management documentation (9 PDFs + 2 MDs)	Complete

GitHub repository structure (src/ modules, Outputs/ subdirectories)	Complete
R1: Feature engineering	Pending
S1: Feature audit (parallel with R1)	Pending
R2: Modeling pipeline	Pending
S5/S6: Figure production	Pending (can start early)